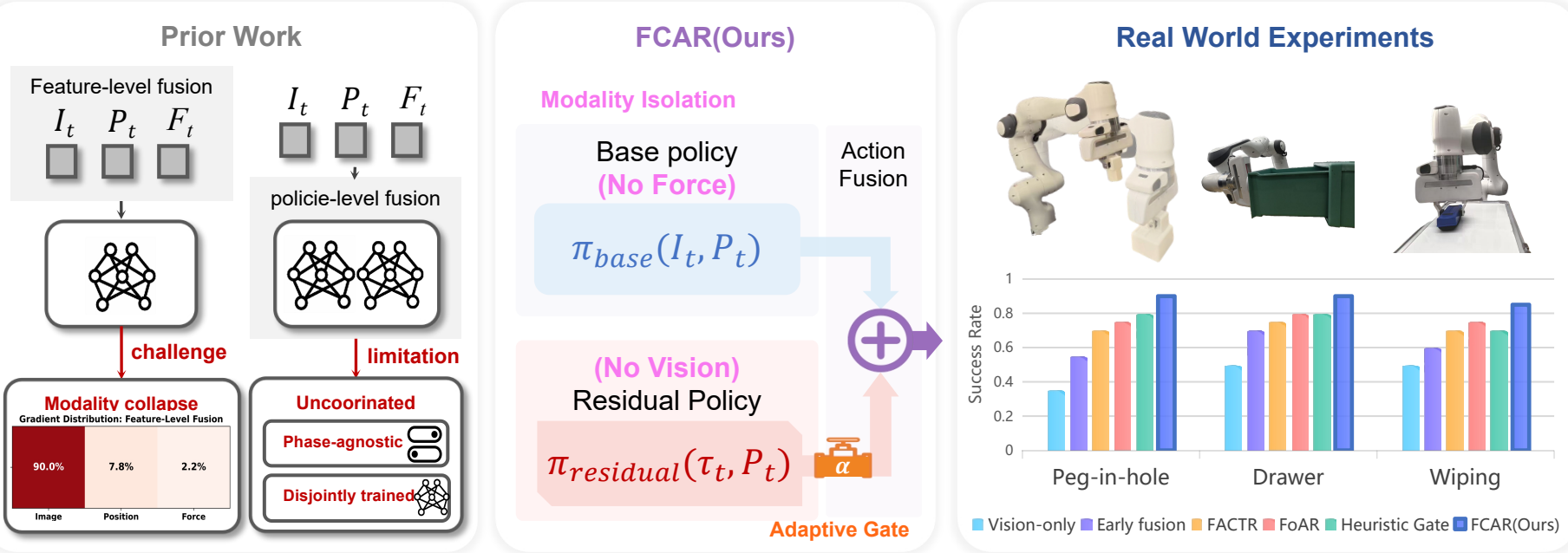
Prior Work
Feature-level fusion
I_t P_t F_t
I_t P_t F_t
policie-level fusion
challenge
limitation
Modality collapse
Gradient Distribution: Feature-Level Fusion
90.0%
7.8%
2.2%
Image
Position
Force
Uncoorinated
Phase-agnostic
Disjointly trained
FCAR(Ours)
Modality Isolation
Base policy
(No Force)
$\pi_{base}(I_t, P_t)$
Action Fusion
(No Vision)
Residual Policy
$\pi_{residual}(\tau_t, P_t)$
α
Adaptive Gate
Real World Experiments
Success Rate
1
0.8
0.6
0.4
0.2
0
Peg-in-hole
Drawer
Wiping
Vision-only
Early fusion
FACTR
FoAR
Heuristic Gate
FCAR(Ours)